



Certificate of Calibration

Ocean Sonics, Ltd.

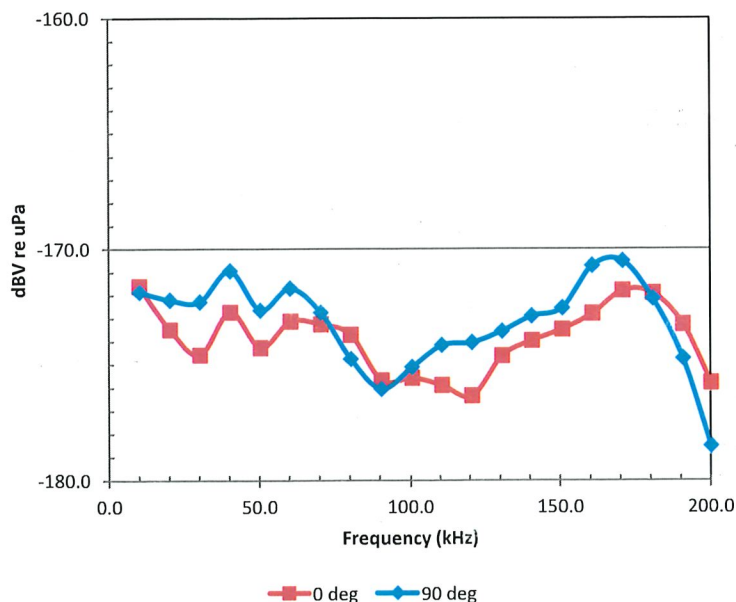
Calibration Certificate Number: **C7787**

Test Result: 10 kHz to 100 kHz: -173.5 ± 2.6
10 kHz to 200 kHz: -174.6 ± 4.0

Model Number SB35-ETH
Serial Number 1273
Manufacture Date 31-Oct-2017
Measurement Date 17-Jan-2024
Certificate Date 17-Jan-2024
Sensitivity @ 26 Hz $-170.3 \text{ dB re V / } \mu\text{Pa}$
Case Type Titanium
Element Manufacturer GeoSpectrum
Element Model M24HF
Element Serial F000851
Preamp Model 04-300435-09
Calibrated By D.Lee
Work Order Number W1234
Test Type RX Sensitivity
Test Procedure Complex RMS
Test Location Tank #3, 1 m
Water Temperature 16 °C

Projector Manufacturer Ocean Sonics
Projector Model TH2-SER-4F
Projector Serial 2225
Measurement Distance 1 m
Output Level 130.7 dB re μPa @ 1 m
Tone Burst 100.0 us / 300 ms
Reference Manufacturer Ocean Sonics
Reference Model RB9-ETH
Reference Serial 6465
Primary Calibration 45111
Preamp Manufacturer Ocean Sonics
Preamp Model 04-300451-01
Preamp Serial Z88
Preamp Gain 36 dB
ADC Manufacturer Ocean Sonics
ADC Model Number 04-300427-01
ADC Serial Number Z87

Receive Sensitivity



Frequency kHz	Sensitivity [dBV re μPa]	
	0 deg	90 deg
10.0	-171.7	-171.9
20.1	-173.5	-172.2
30.1	-174.6	-172.3
40.2	-172.8	-171.0
50.2	-174.3	-172.7
60.2	-173.2	-171.7
70.3	-173.3	-172.8
80.3	-173.8	-174.8
90.4	-175.7	-176.1
100.4	-175.6	-175.2
110.4	-175.9	-174.2
120.5	-176.4	-174.1
130.5	-174.7	-173.6
140.5	-174.0	-172.9
150.6	-173.5	-172.6
160.6	-172.8	-170.8
170.7	-171.8	-170.6
180.7	-172.0	-172.2
190.7	-173.3	-174.8
200.0	-175.8	-178.6